

# 15 — Implementation Phase

## Purpose

The purpose of this document is to define the implementation process of Mosaica.

This phase transforms the approved architecture and development preparation decisions into a working software system.

The objective is to create Mosaica through controlled implementation steps while preserving architectural consistency.

---

## Implementation Scope

This phase includes:

- development environment activation,
  - project creation,
  - application initialization,
  - first working version,
  - implementation validation.
- 

## Implementation Position

Implementation begins after architecture approval and development preparation completion.

Approved Architecture

↓

Development Preparation

↓

Implementation

↓

Validation

---

## Implementation Principles

## Principle 1 — Build Working Software Early

The goal is to create a functioning system as soon as possible.

---

## Principle 2 — Follow Approved Architecture

Implementation decisions must respect established architecture.

---

## Principle 3 — Start Small

The first version should prove the core experience before expansion.

---

## Principle 4 — Validate Continuously

Each development step should be tested before continuing.

---

## Initial Implementation Goal

The first milestone is:

Mosaica v0.1

The first version should provide:

- running application,
  - basic project structure,
  - first database connection,
  - initial user experience foundation.
- 

## Implementation Flow

Environment

↓

Project Creation

↓

Application Setup

↓

First Running Version

↓

Core Features

---

## Module Structure

15 – Implementation Phase

15.01 Environment Installation

15.02 Project Creation

15.03 Initial Application Setup

15.04 First Running Version

15.05 Implementation Review

## 15.01 — Environment Installation

### Purpose

The purpose of this section is to define and prepare the local development environment required for Mosaica implementation.

This section establishes the required software tools and configurations needed to create, run and maintain the application.

The objective is to create a reliable and free development environment.

---

## Installation Principles

---

### Principle 1 — Minimum Required Setup

Only tools required for initial development shall be installed.

---

## Principle 2 — Free Tools First

The initial development environment shall use free tools.

---

## Principle 3 — Local Development First

Mosaica shall initially run on the developer's computer without requiring external infrastructure.

---

## Required Development Tools

The initial environment requires:

1. Visual Studio Code
  2. Node.js
  3. Git
  4. GitHub Account
  5. Modern Web Browser
- 

## 1. Visual Studio Code

### Purpose

Primary development editor.

---

### Usage

Visual Studio Code will be used for:

- writing TypeScript code,
- managing project files,
- running commands,
- reviewing changes.

---

## Cost

Free.

---

## 2. Node.js

### Purpose

Provides the runtime environment required by Next.js.

---

### Usage

Node.js enables:

- running the development server,
  - installing project packages,
  - executing development commands.
- 

## Cost

Free.

---

## 3. Git

### Purpose

Version control system.

---

### Usage

Git provides:

- change tracking,

- commit history,
  - safe development workflow.
- 

## **Cost**

Free.

---

# **4. GitHub Account**

## **Purpose**

Remote repository management.

---

## **Usage**

GitHub provides:

- code backup,
  - repository hosting,
  - version history storage.
- 

## **Cost**

Free plan is sufficient.

---

# **5. Web Browser**

## **Purpose**

Application testing.

---

Recommended:

- Chrome,
- Edge,
- Firefox.

---

## Installation Order

The installation order:

Visual Studio Code

↓

Node.js

↓

Git

↓

GitHub Setup

↓

Mosaica Project Creation

---

## Excluded Tools

The following are not required now:

Docker

Cloud Services

Database Hosting

AI Tools

Paid Software

---

## Environment Validation

After installation, the environment should confirm:

Code Editor Ready

+

Runtime Ready

+

Version Control Ready

+

Project Hosting Ready

---

# Environment Decision

## Decision

Mosaica development will begin using a local free development environment based on VS Code, Node.js and Git.

---

## Rationale

This approach provides:

- zero initial cost,
- simple setup,
- full development capability,
- future expansion possibility.

## 15.02 — Project Creation

### Purpose

The purpose of this section is to define the creation process of the initial Mosaica application project.

This section establishes the first executable version of the software project based on the approved technology stack.

The objective is to create a clean and maintainable starting point for Mosaica development.

---

## Project Creation Principles

---

# Principle 1 — Clean Initial Setup

## Definition

The project should begin from a clean and official framework structure.

---

## Application

Avoid manually creating project files before initialization.

The framework should create the required foundation.

---

# Principle 2 — Follow Selected Technology Stack

The project must use:

Framework:  
Next.js

Language:  
TypeScript

Styling:  
Tailwind CSS

Database Preparation:  
Prisma

---

# Principle 3 — Local First Development

The initial project shall run locally on the developer's computer.

---

## Initial Project Location

The recommended location:

Documents

└─ Development

# Project Creation Process

The initial creation flow:

Create Development Folder

↓

Open Terminal

↓

Create Next.js Project

↓

Install Dependencies

↓

Start Development Server

↓

Verify Application

---

# Initial Project Configuration

The project should include:

TypeScript



ESLint



Tailwind CSS



App Router



src Directory



---

# Project Naming

Project name:

mosaica

Naming rules:

- lowercase,
- simple,
- consistent with repository name.

---

## First Project Goal

The first successful result:

Mosaica application runs locally

Expected result:

Browser displays:

`http://localhost:3000`

---

## Initial Project State

After creation:

```
mosaica/  
├─ app/  
├─ public/  
├─ src/  
├─ package.json  
└─ configuration files
```

---

## Project Creation Decision

### Decision

Mosaica will begin as a clean Next.js TypeScript project created through the official initialization process.

---

## Rationale

This provides:

- reliable foundation,
- standard structure,
- easier maintenance,
- future scalability.

## 15.03 — Initial Application Setup

### Purpose

The purpose of this section is to define the initial configuration and preparation of the Mosaica application after project creation.

This section transforms the default framework installation into a clean Mosaica development foundation.

The objective is to remove unnecessary starter elements and prepare the application structure for feature development.

---

## Initial Setup Principles

### Principle 1 — Remove Framework Defaults

The initial template should be replaced with Mosaica-specific structure.

---

### Principle 2 — Preserve Working Foundation

The application setup should maintain the official Next.js structure.

---

### Principle 3 — Prepare Before Feature Development

Core features should be added only after the application foundation is clean.

---

## Initial Setup Tasks

Remove Default Content

↓

Configure Application Identity

↓

Create Initial Structure

↓

Verify Application

---

## Initial Cleanup Scope

The following default elements will be replaced:

- default Next.js homepage,
  - Vercel branding,
  - starter content.
- 

## Initial Mosaica Foundation

After setup, the application should represent:

Mosaica

Personal Cultural Archive System

## 15.04 — First Running Version

### Purpose

The purpose of this section is to define and validate the first running version of Mosaica.

This version represents the first functional state of the application after environment setup and initial configuration.

The objective is to confirm that the development foundation works correctly before implementing core features.

---

# First Version Definition

## Mosaica v0.1

Mosaica v0.1 represents:

A running local application  
with  
a prepared development foundation

---

## Included in v0.1

The first version includes:

Application Startup  
↓  
Basic Interface  
↓  
Project Structure  
↓  
Development Environment

---

## Not Included in v0.1

The following are intentionally excluded:

Database Features  
User Accounts  
Archive System  
Collections  
AI Features  
Advanced Design

---

# First Version Validation

The following requirements must be satisfied:

---

## Application Runs

Status:

✓ Confirmed

Result:

`localhost:3000`

opens successfully.

---

## Code Updates Work

Status:

✓ Confirmed

Result:

Changes to files are reflected automatically through Next.js development mode.

---

## TypeScript Environment Works

Status:

✓ Confirmed

Result:

Project successfully runs with TypeScript configuration.

---

## Project Structure Exists

Status:

✓ Confirmed

Result:

The initial application structure has been created.

---

## Current Mosaica State

Mosaica v0.1

Frontend Foundation



Development Environment



Local Execution



Core Features

☐ Future

---

## First Running Version Decision

### Decision

Mosaica has reached its first successful running version.

---

### Rationale

The application foundation is operational and ready for feature development.

---

## 15.04 Status Preparation

After this validation, the next step will be:

Implementation Review

The review will confirm:

- whether the first version is stable,
- whether the project is ready for actual feature development.

## 15.05 — Implementation Review

### Purpose

The purpose of this review is to validate the successful completion of the initial Mosaica implementation phase.

This review confirms that the application foundation is operational and ready for feature development.

The objective is to ensure that the transition from preparation into active product development can proceed safely.

---

### Review Scope

The following areas are evaluated:

Environment Setup

Project Creation

Application Configuration

Local Execution

Development Readiness

---

### Validation Results

---

# 1. Development Environment

## Status: Approved

Confirmed:

Visual Studio Code  
✓

Node.js  
✓

npm  
✓

Git  
✓

The required development environment is operational.

---

# 2. Project Creation

## Status: Approved

Confirmed:

Next.js Project

TypeScript Setup

Package Management

Project Files

The Mosaica project has been successfully created.

---

# 3. Application Configuration

## Status: Approved

Confirmed:

- default Next.js page replaced,
- Mosaica identity added,

- application metadata configured.

Current application state:

Mosaica

Kişisel kültür arşivi sistemi

---

## 4. Local Execution

### Status: Approved

Confirmed:

Application successfully runs at:

`http://localhost:3000`

Development server works correctly.

---

## 5. Development Workflow

### Status: Approved

Confirmed:

The project can now follow:

Change Code

↓

Save

↓

Automatic Update

↓

Test

workflow.

---

# Implementation Phase Assessment

## Current Strengths

Mosaica currently has:

Working Development Environment

+

Created Application

+

Controlled Structure

+

Version Control Foundation

---

## Current Limitations

The application does not yet contain:

Database

User System

Archive Features

Content Management

Advanced Interface

These belong to future development phases.

---

# Implementation Decision

## Decision

The initial implementation phase has been successfully completed.

Mosaica is ready to move from technical setup into product feature development.

---

# Phase Status

15 — Implementation Phase

Status: APPROVED

State: Ready for Feature Development